

Lab-1 Software Engineering

Name: Kshitij G Shettigar

SRN: PES1UG24CS240

Problem Statement: Community Tool & Equipment Library

Overview: A neighbourhood resource-sharing library where members check out power tools, camping gear, and lawnmowers, track deposit calculations, manage return dates, and log damage assessments.

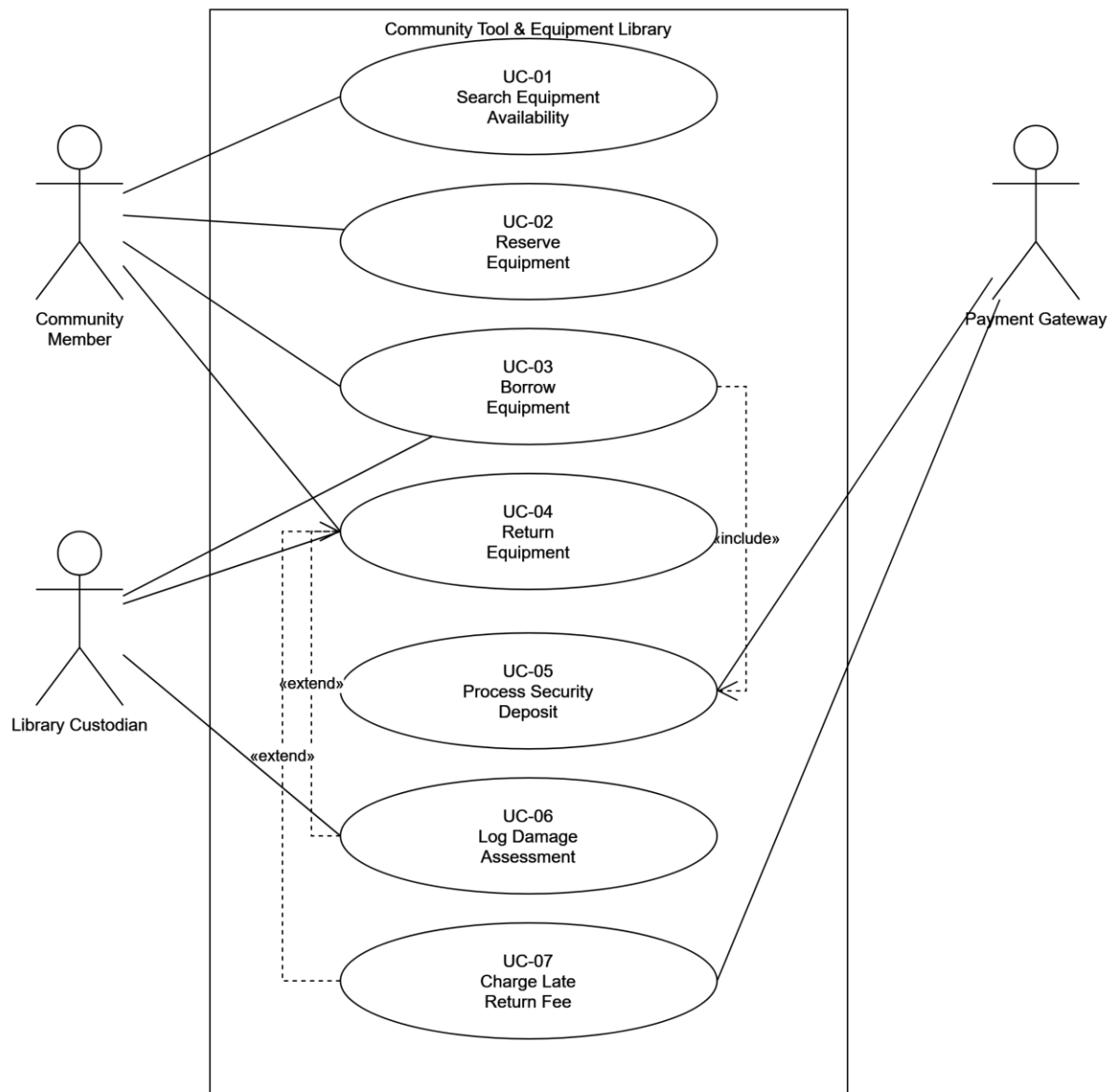
Target Stakeholders / Actors: Community Member, Library Custodian (Payment Gateway is also an Actor)

Deliverable 1 : FRs and NFRs

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall calculate the required security deposit for each item from its tool category, enforce that category's maximum loan duration, and compute a late return fee for each day an item is held beyond its due date.	High	Pass: A Category-C item (deposit Rs.2000, max loan 3 days) holds Rs. 2000 at checkout and falls due in 3 days; a 2-day-late return is charged 2x the category daily rate. Fail: A zero deposit is held, a loan exceeds the category maximum, or a late return yields no fee.	Deposits and duration caps are the library's only protection against loss of high-value equipment.	
FR-002	Functional	The system shall allow a Community Member to search the equipment catalogue by category, keyword, or required date, and shall show for each result its status (Available / Reserved / On Loan / Under Repair) and the depot locker holding it.	High	Pass: A search for 'lawnmower' returns only lawnmowers, each tagged with a status and a locker ID, and an item checked out 5 seconds earlier already shows On Loan. Fail: A result shows no locker, or an item already on loan is listed Available.	Members travel to a physical depot to collect; a wrong status or locker ID sends them to the wrong place.	

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-003	Functional	The system shall allow a Community Member to reserve an available item for a pickup window of at most 24 hours, and shall release the reservation automatically if the item is not collected before that window closes.	Medium	Pass: A reservation marks the item Reserved and blocks other members from reserving it; an uncollected reservation returns to Available within 5 minutes of the window closing. Fail: Two members hold reservations on one item, or an expired reservation still blocks it an hour later.	Holds prevent wasted trips, but holds that never expire make scarce tools look permanently unavailable.	
FR-004	Functional	The system shall verify, before confirming any checkout, that the Community Member has no overdue item and no unpaid damage or late fee, and shall block checkout of high-value equipment when either condition fails.	High	Pass: A member holding one overdue drill is refused a chainsaw and is shown the blocking reason. Fail: A member with overdue tools borrows additional high-value equipment.	Unenforced eligibility lets one member accumulate the inventory the rest of the neighbourhood is waiting on.	
FR-005	Functional	The system shall record the condition of every returned item, allow the Library Custodian to log a damage assessment with severity and repair cost, deduct that cost and any late fee from the held deposit, and release the balance to the member.	High	Pass: A return carrying Rs.500 damage and Rs.120 late fee against a Rs.2000 deposit releases Rs.1380 with an itemised breakdown. Fail: The deposit is released in full despite logged damage, or the deduction does not equal damage + late fee.	Deposit settlement is the step members dispute; an itemised, auditable breakdown is what makes the charge defensible.	
NFR-001	Nonfunc.	The equipment inventory search shall return real-time physical availability across all neighbourhood depot lockers within 2 s at the 95th percentile while sustaining 200 concurrent member sessions and shall expose locker contents only to authenticated members.	High	Pass: A load test at 200 concurrent users records p95 search latency of at most 2 s, locker status no more than 10 s stale, and HTTP 401 for an unauthenticated locker request. Fail: p95 above 2 s, staler data, or unauthenticated access succeeds.	Slow or stale availability sends members to empty lockers, and locker contents reveal where valuable tools physically sit.	
NFR-002	Nonfunc.	The system shall process all deposit holds and fee charges through the external payment gateway using tokenised references, shall store no card data, and shall retain an immutable audit record of every hold, deduction and release for 12 months.	High	Pass: A database scan after 100 transactions finds no card number or CVV, only gateway tokens, and every transaction has a timestamped audit entry the application cannot alter. Fail: Any card field is recoverable, or a deduction has no matching audit entry.	Money moves between neighbours rather than through a merchant, so the library must be able to prove what it charged and cannot justify holding card data.	

Deliverable 2 : UML Use-Case Diagram



Deliverable 3 : Use-Case Specification

Use Case ID	UC-03
Use Case Name	Borrow Equipment
Primary Actor	Community Member
Secondary Actors	Library Custodian, Payment Gateway
Related Requirements	FR-001, FR-002, FR-004, NFR-002
Description	A Community Member borrows an item of equipment from the library. The system checks the member's eligibility, calculates and holds a security deposit, and records the loan against a due date.
Preconditions	1. The Community Member is registered and authenticated on the system. 2. The member has no overdue item and no unpaid damage or late fee (FR-004). 3. The requested item's status is Available, or Reserved by that same member. 4. The member has a valid deposit payment method registered with the Payment Gateway.
Postconditions	1. The item is marked On Loan against the member, with a due date set from its tool category's maximum loan duration. 2. A security deposit for that category is held against the member's payment method. 3. The depot locker inventory is updated so the item no longer appears as available. 4. A timestamped audit entry records both the loan and the deposit hold (NFR-002).
Main Success Scenario	1. The Community Member searches the catalogue for the required equipment. 2. The system displays matching items with their current status and depot locker (FR-002). 3. The member selects an available item and requests to borrow it. 4. The system verifies the member's eligibility - no overdue items and no unpaid fees (FR-004). 5. The system calculates the security deposit and the loan duration limit for the item's tool category (FR-001). 6. The system requests a deposit hold from the Payment Gateway. [includes UC-05 Process Security Deposit] 7. The Payment Gateway authorises the hold and returns a tokenised reference. 8. The system records the loan, sets the due date, and marks the item On Loan. 9. The Library Custodian releases the item from the depot locker to the member. 10. The system issues a confirmation showing the due date and the deposit amount held. The use case ends successfully.
Alternate Flow	6a. Deposit authorisation declined 6a1. The Payment Gateway rejects the deposit hold and returns a decline reason. 6a2. The system notifies the member and prompts for an alternative payment method. 6a3. The member selects a different method and the flow resumes at step 6. 6a4. If authorisation fails again after 2 attempts, the system cancels the borrow request, returns the item to Available, and notifies the member. The use case ends in failure.

